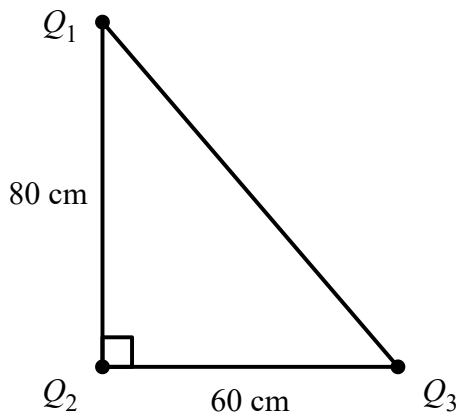
STPM 2019 PHYSICS 960/2

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $N_A = 6.02 \times 10^{23}$ mol^{-1}, $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ N m^2 C^{-2}, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}. *Q14: statements A and C are both untrue, so A or C should be accepted.*

Section A [15 marks]

1. Three charged particles, $Q_1 = -45$ nC, $Q_2 = +32$ nC and $Q_3 = -56$ nC, are placed at the vertices of a triangle as shown in the diagram below.



If Q_1 and Q_3 are 80 cm and 60 cm from Q_2 respectively, what is the magnitude of the resultant force which acts on Q_2?

A 2.0×10^{-5} N B 4.5×10^{-5} N C 4.9×10^{-5} N D 6.5×10^{-5} N

Answer: C

$$F_{12} = \frac{(9.0 \times 10^9)(45 \times 10^{-9})(32 \times 10^{-9})}{(0.80)^2} = 2.0 \times 10^{-5} \text{ N (towards } Q_1)$$

$$F_{32} = \frac{(9.0 \times 10^9)(56 \times 10^{-9})(32 \times 10^{-9})}{(0.60)^2} = 4.5 \times 10^{-5} \text{ N (towards } Q_3)$$

The two forces are perpendicular: $F = \sqrt{(2.0^2 + 4.5^2)} \times 10^{-5} = 4.9 \times 10^{-5}$ N

Explanation: Both forces are attractions, along the two perpendicular sides.

2. The distance between two point charges, $+2q$ and $+3q$, is R. What is the electric potential at the midpoint between the two charges?

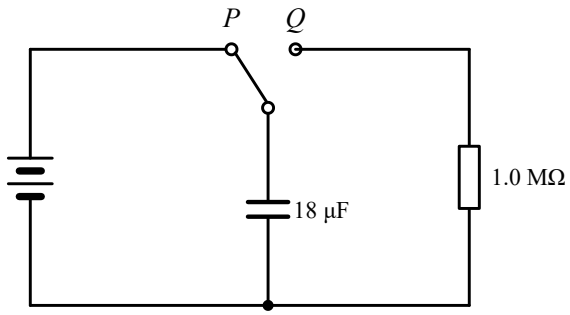
A $\frac{q}{\pi\varepsilon_0 R^2}$ B $\frac{5q}{\pi\varepsilon_0 R^2}$ C $\frac{q}{4\pi\varepsilon_0 R}$ D $\frac{5q}{2\pi\varepsilon_0 R}$

Answer: D

$$V = \frac{2q + 3q}{4\pi\varepsilon_0 (R/2)} = \frac{10q}{4\pi\varepsilon_0 R} = \frac{5q}{2\pi\varepsilon_0 R}$$

Explanation: Potential is a scalar: just add.

3. A capacitor of capacitance 18 μF and a resistor of resistance 1.0 MΩ are connected to a battery of e.m.f. 200 V as shown in the circuit below.



Initially, the switch is connected to *P* to fully charge the capacitor. Then the switch is connected to *Q* to discharge the capacitor. What is the time taken for the charge in the capacitor to decrease to 37% of the initial value?

A 18 s B 12 s C 8.3 s D 6.7 s

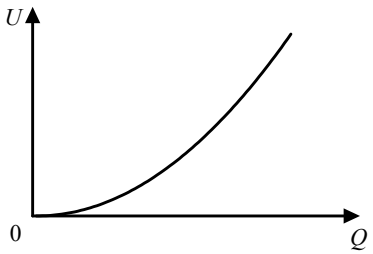
Answer: A

37% of the initial charge remains after one time constant: $\tau = RC = (1.0 \times 10^{6})(18 \times 10^{-6}) = 18$ s

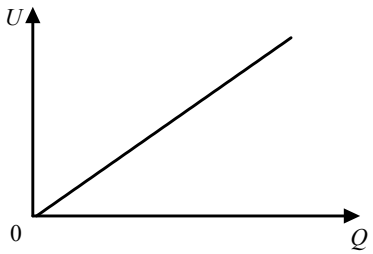
Explanation: $e^{-1} = 0.37$.

4. Which graph shows the variation of energy stored, *U*, in the capacitor with charge *Q* during the charging process?

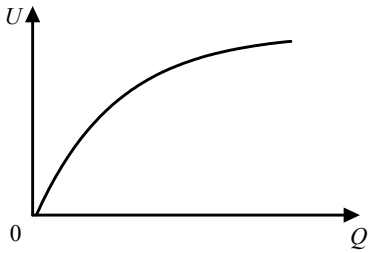
A



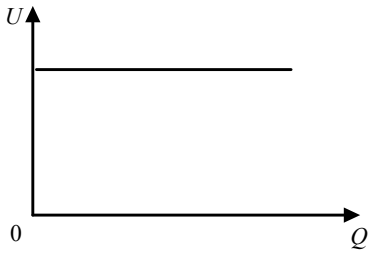
B



C



D



Answer: A

$$U = \frac{Q^2}{2C} \propto Q^2\text{: a parabola through the origin (A)}$$

Explanation: It is the V–Q graph that is a straight line.

5. The density of copper is 8920 kg m^{-3} and its relative atomic mass is 63.5. If a copper wire carries a current with current density of 1.67×10^6 A m^{-2}, what is the drift velocity of the electrons in the wire?

A 7.12×10^{-3} m s^{-1}
B 1.23×10^{-4} m s^{-1}
C 4.19×10^{-5} m s^{-1}
D 1.51×10^{-7} m s^{-1}

Answer: B

$$n = \frac{\rho N_A}{M} = \frac{(8920)(6.02 \times 10^{23})}{63.5 \times 10^{-3}} = 8.46 \times 10^{28}\ \text{m}^{-3}$$

$$v = \frac{J}{ne} = \frac{1.67 \times 10^6}{(8.46 \times 10^{28})(1.60 \times 10^{-19})} = 1.23 \times 10^{-4}\ \text{m s}^{-1}$$

Explanation: One free electron per copper atom.

6. Two cylindrical resistors are made of copper and aluminium respectively. The cross-sectional area and electrical conductivity of the copper are four times the cross-sectional area and conductivity of the aluminium. If the lengths of both resistors are the same, what is the ratio of resistance of the aluminium resistor to resistance of the copper resistor?

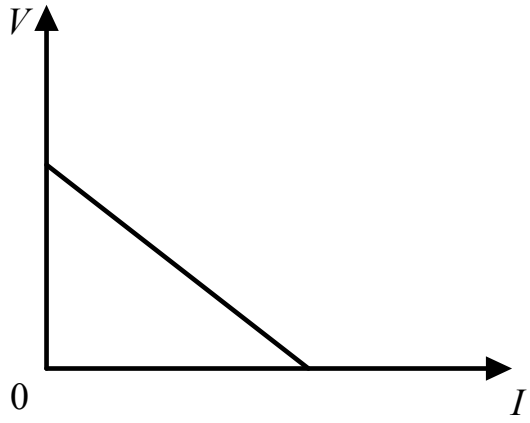
A $\frac{1}{16}$ B $\frac{1}{4}$ C 4 D 16

Answer: D

$$R = \frac{l}{\sigma A}\text{:}\ \frac{R_{\text{Al}}}{R_{\text{Cu}}} = \frac{\sigma_{\text{Cu}} A_{\text{Cu}}}{\sigma_{\text{Al}} A_{\text{Al}}} = 4 \times 4 = 16$$

Explanation: Higher conductivity and larger area both reduce resistance.

7. The variation of terminal potential difference V of a battery with current I which flows in the circuit is shown in the graph below.



The internal resistance of the battery can be determined from the

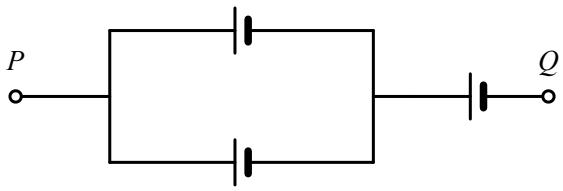
A x-intercept of the graph
B y-intercept of the graph
C gradient of the graph
D area under the graph

Answer: C

$V = \varepsilon - Ir$: the gradient of the graph is $-r$

Explanation: The y-intercept gives the e.m.f.

8. Three identical dry cells, each of e.m.f. 1.5 V and internal resistance 0.90 Ω, are connected as shown in the diagram below.



The combination of the dry cells is equivalent to one dry cell of e.m.f. ε and internal resistance r. What are the values of ε and r?

	ε	r
A	3.0 V	2.70 Ω
B	3.0 V	1.35 Ω
C	1.5 V	1.35 Ω
D	1.5 V	0.30 Ω

Answer: B

Two cells in parallel: e.m.f. 1.5 V, internal resistance 0.90/2 = 0.45 Ω.

In series with the third cell: ε = 3.0 V, r = 0.45 + 0.90 = 1.35 Ω.

Explanation: Identical cells in parallel keep the same e.m.f.

9. Which condition does **not** produce a magnetic field?

A The moving charge
B The moving bar magnet
C The constant electric field
D The constant electric current

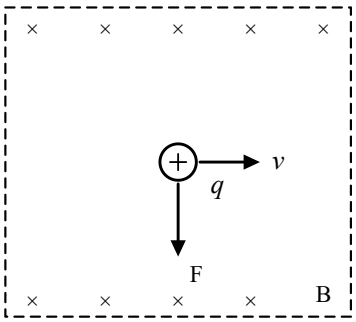
Answer: C

A magnetic field is produced by moving charges (currents) and magnets; a constant electric field alone does not produce one (C).

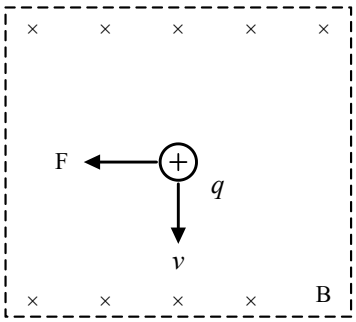
Explanation: Only a changing electric field produces a magnetic field.

10. When a point charge with a magnitude q moves with a velocity $\boldsymbol{v}$ in a magnetic field $\boldsymbol{B}$, a magnetic force $\boldsymbol{F}$ acts on the charge. Which diagram shows the correct orientation of $\boldsymbol{F}$ relative to $\boldsymbol{B}$ and $\boldsymbol{v}$?

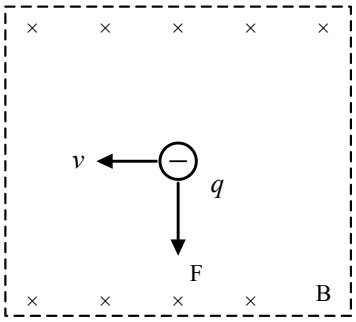
A



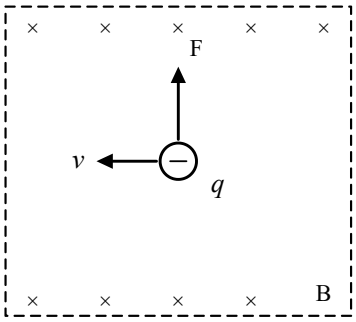
B



C



D



Answer: D

B is into the page. For $-q$ moving to the left, $\mathbf{v} \times \mathbf{B}$ points down, so the force on a negative charge points up (D).

Explanation: Check each option with $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$, remembering the sign of q.

11. A conducting plate of thickness 0.50 mm is placed perpendicularly to a magnetic field of 1.2 T. What is the Hall voltage generated when a current of 3.4 A flows through the conductor if there are 8.0×10^{28} electrons per m^3?

A 2.2 μV B 0.64 μV C 0.32 μV D 0.13 μV

Answer: B

$$V_H = \frac{BI}{net} = \frac{(1.2)(3.4)}{(8.0 \times 10^{28})(1.60 \times 10^{-19})(0.50 \times 10^{-3})} = 6.4 \times 10^{-7}\ \text{V} = 0.64\ \mu\text{V}$$

Explanation: Hall voltages in metals are tiny because n is very large.

12. The current in a coil decreases uniformly from 3.6 A to 1.4 A in 0.25 s. What is the e.m.f. induced in the coil if the inductance of the coil is 4.4 mH?

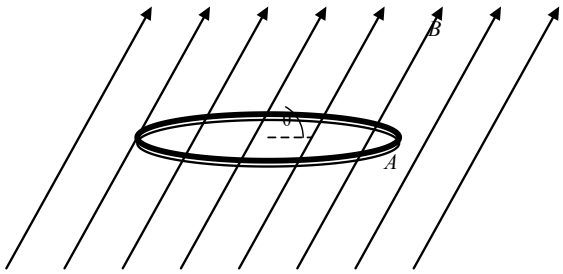
A 1.3×10^{-1} V B 3.9×10^{-2} V C 2.4×10^{-3} V D 5.0×10^{-4} V

Answer: B

$$\varepsilon = L\frac{\Delta I}{\Delta t} = (4.4 \times 10^{-3})\frac{3.6 - 1.4}{0.25} = 3.9 \times 10^{-2}\ \text{V}$$

Explanation: The rate of change of current is 8.8 A s^{-1}.

13. The plane of a circular coil of N turns, with an area A, is placed at an angle θ in a uniform magnetic field as shown in the diagram below.



What is the magnetic flux linkage through the coil, if the magnetic flux density is B?

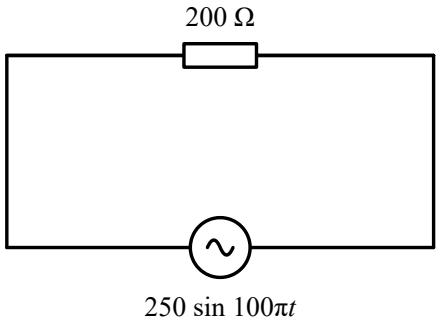
A BA B NBA C $NBA \cos\theta$ D $NBA \sin\theta$

Answer: D

θ is measured between the plane of the coil and **B**, so the component of **B** normal to the plane is $B \sin\theta$: $N\Phi = NBA \sin\theta$

Explanation: If θ were measured from the normal, it would be $\cos\theta$.

14. A resistor of resistance 200 Ω is connected in series with an a.c. source $V = 250 \sin 100\pi t$, as shown in the circuit below.



Which is **not** true about the circuit?

A The maximum current of the circuit is 0.625 A.
B The r.m.s. current of the circuit is 0.884 A.
C The maximum output voltage of the circuit is 177 V.
D The maximum output power of the circuit is 313 W.

Answer: A

$$I_0 = \frac{250}{200} = 1.25 \text{ A, so A (0.625 A) is not true; } I_{rms} = \frac{1.25}{\sqrt{2}} = 0.884 \text{ A (B true)}$$

The maximum voltage is 250 V, so C (177 V, the r.m.s. value) is also not true; the peak power $250^2/200 = 313$ W (D true).

Explanation: Two statements are untrue, so A or C should be accepted.

15. A pure inductor of self inductance 0.12 H and a pure resistor of resistance 40 Ω are connected in series with a sinusoidal voltage supply of frequency 50 Hz. What is the r.m.s. current in the circuit if the r.m.s. voltage is 230 V?

A 0.14 A B 3.0 A C 4.2 A D 5.8 A

Answer: C

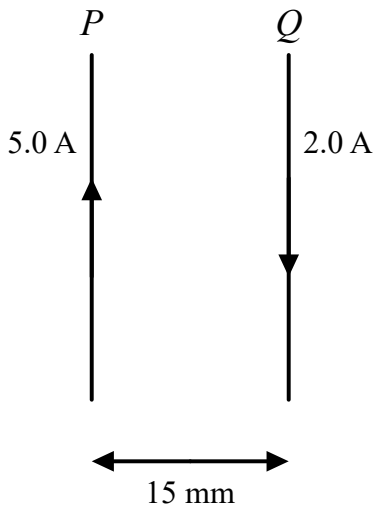
$$X_L = 2\pi(50)(0.12) = 37.7 \ \Omega; \ Z = \sqrt{(40^2 + 37.7^2)} = 55.0 \ \Omega$$

$$I_{rms} = \frac{230}{55.0} = 4.2 \text{ A}$$

Explanation: Resistance and reactance add as phasors.

Section B [15 marks]

16. Two long parallel wires, P and Q, are separated by a distance of 15 mm, carrying currents of 5.0 A and 2.0 A respectively in opposite directions, as shown in the diagram below.



(a) Calculate the resultant magnetic field at the midway point between P and Q. [4]

Answer

Each wire is 7.5 mm from the midpoint	1
$B_P = \frac{\mu_0(5.0)}{2\pi(0.0075)} = 1.33 \times 10^{-4}$ T; $B_Q = \frac{\mu_0(2.0)}{2\pi(0.0075)} = 5.3 \times 10^{-5}$ T	1
The currents are opposite, so both fields at the midpoint point into the page and add	1
$B = 1.87 \times 10^{-4}$ T, into the page	1
	4

Explanation: Right-hand grip rule for each wire.

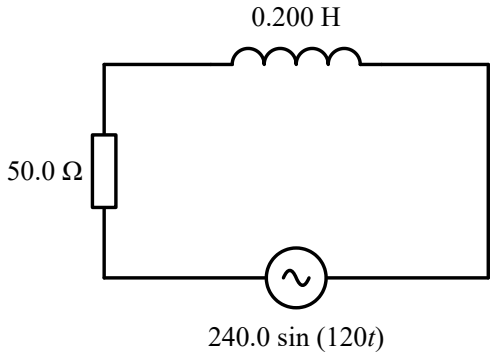
(b) Explain the direction of movements of P and Q. [3]

Answer

Each wire lies in the magnetic field of the other and experiences a force BIl	1
For opposite currents the forces are repulsive	1
P moves to the left and Q to the right, with equal and opposite forces (Newton's third law)	1
	3
Total for question	**7**

Explanation: Like currents attract; unlike currents repel.

17. A sinusoidal voltage $V = 240.0 \sin (120t)$ is supplied to a circuit which consists of a resistor of resistance 50.0 Ω and an inductor of inductance 0.200 H as shown in the diagram below.



(a) Calculate the impedance of the circuit. [3]

Answer

$X_L = \omega L = (120)(0.200) = 24.0\ \Omega$ — 1

$Z = \sqrt{(50.0^2 + 24.0^2)}$ — 1

$Z = 55.5\ \Omega$ — 1

3

Explanation: $\omega = 120$ rad s^{-1} from the equation.

(b) Determine the maximum voltage across the inductor. [2]

Answer

$I_0 = \frac{240.0}{55.5} = 4.33$ A — 1

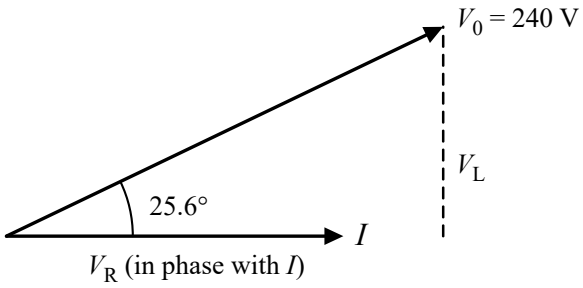
$V_{L,0} = I_0 X_L = (4.33)(24.0) = 104$ V — 1

2

Explanation: The maximum voltages across R and L occur at different times.

(c) Calculate the phase angle and sketch a phasor diagram of the voltage supply and the current in the circuit. [3]

Answer



$\tan \varphi = \frac{X_L}{R} = \frac{24.0}{50.0}$ 1

$\varphi = 25.6°$ 1

Phasor diagram with the supply voltage leading the current by 25.6° 1

3

Total for question **8**

Explanation: In an *RL* circuit the current lags the voltage.

Section C [30 marks]

18. (a) Two charged objects 12 cm apart repel each other with a force of 5.0 N. If the distance is reduced to 4.0 cm, determine the force experienced by the two objects. [3]

Answer

$F \propto \frac{1}{r^2}$ 1

$F = 5.0 \times (\frac{12}{4.0})^2$ 1

$F = 45$ N 1

3

Explanation: Reducing the distance to one third multiplies the force by 9.

(b) A solid metal sphere with a radius of r_a is located in a metal spherical shell with a radius of r_b. There is a charge $+q$ on the solid metal sphere and a charge of $-q$ on the surface of the spherical shell.

(i) Determine the electric potential for $r < r_a$ and $r_a < r < r_b$. [3]

Answer

$r < r_a$ (inside the conductor): V is constant, equal to its surface value	1
$V = \frac{q}{4\pi\varepsilon_0}(\frac{1}{r_a} - \frac{1}{r_b})$	1
$r_a < r < r_b$: $V = \frac{q}{4\pi\varepsilon_0}(\frac{1}{r} - \frac{1}{r_b})$	1
	3

Explanation: The shell's $-q$ adds $-q/4\pi\varepsilon_0 r_b$ everywhere inside it.

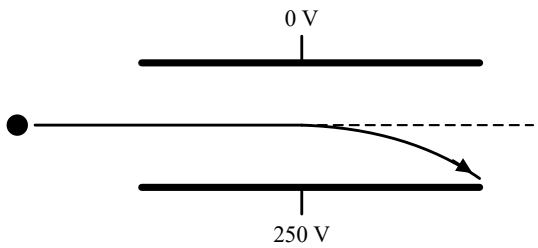
(ii) If the metal spherical shell with the radius r_b is removed, determine the electric field at any point outside the solid sphere using Gauss's law. [4]

Answer

Choose a spherical Gaussian surface of radius $r > r_a$, concentric with the sphere	1
By symmetry E is radial and constant on it: flux = $E(4\pi r^2)$	1
Gauss: $E(4\pi r^2) = \frac{q}{\varepsilon_0}$	1
$E = \frac{q}{4\pi\varepsilon_0 r^2}$, radially outwards	1
	4

Explanation: Outside, the sphere acts like a point charge at its centre.

(c) Two oppositely charged horizontal plates 1.3 cm apart have a potential difference of 250 V. An electron moves into the region at the point equidistant between the plates as shown in the diagram below.



(i) Determine the electric field between the plates. [2]

Answer

$E = \frac{V}{d} = \frac{250}{0.013}$ 1

$E = 1.9 \times 10^4 \text{ V m}^{-1}$ 1

2

Explanation: Directed from the 250 V plate towards the 0 V plate.

(ii) Determine the work done for the electron to emerge at one end of the plate. [3]

Answer

The electron enters at the mid-plane, where $V = 125$ V 1

It is attracted to the 250 V plate and emerges at its edge: $\Delta V = 250 - 125 = 125$ V 1

$W = e\Delta V = (1.60 \times 10^{-19})(125) = 2.0 \times 10^{-17}$ J 1

3

Total for question **15**

Explanation: The work is done by the electric field and appears as kinetic energy.

19. (a) Define *capacitance* of a capacitor. [1]

Answer

The ratio of the charge on either plate to the p.d. between the plates, $C = \frac{Q}{V}$ 1

1

Explanation: Unit: farad.

(b) Describe the mechanism of charging a parallel plate capacitor. [4]

Answer

When connected to a battery, electrons flow from the plate joined to the positive terminal, through the battery, to the other plate 1

One plate becomes positively charged and the other equally negatively charged 1

The p.d. across the plates rises as charge builds up, so the charging current decreases 1

Charging stops when the p.d. across the plates equals the e.m.f. 1

4

Explanation: No charge crosses the gap between the plates.

(c) What is the effect on capacitance if an insulating material is inserted between the parallel plate capacitor? Explain how the effect happens. [4]

Answer

The capacitance increases (by a factor ε_r) 1

The insulator's molecules are polarised by the field between the plates 1

Induced surface charges on the insulator produce a field opposing the applied field, reducing the field and the p.d. 1

For the same charge V is smaller, so $C = \frac{Q}{V}$ increases (or more charge is stored at the same p.d.) 1

4

Explanation: The insulator is a dielectric.

(d) A parallel plate capacitor of area 144 cm^2 is charged by a battery of 6.0 V. The separation between the plates is 10.0 mm and the space between the plates is filled by mica of dielectric constant 5.4.

(i) Determine the capacitance of the capacitor. [2]

Answer

$C = \frac{\varepsilon_r \varepsilon_0 A}{d} = \frac{(5.4)(8.85 \times 10^{-12})(144 \times 10^{-4})}{10.0 \times 10^{-3}}$ 1

$C = 6.88 \times 10^{-11}$ F 1

2

Explanation: 144 cm^2 = 0.0144 m^2.

(ii) Calculate the electric field between the plates. [2]

Answer

$E = \frac{V}{d} = \frac{6.0}{10.0 \times 10^{-3}}$ 1

$E = 600$ V m^{-1} 1

2

Explanation: The battery stays connected, so V = 6.0 V.

(iii) Calculate the energy stored in the capacitor. [2]

Answer

$U = \frac{1}{2}CV^2 = \frac{1}{2}(6.88 \times 10^{-11})(6.0)^2$ 1

$U = 1.24 \times 10^{-9}$ J 1

2

Total for question **15**

Explanation: Equivalently ½QV.

20. (a) Describe a potential divider as a source of variable voltage. [2]

Answer

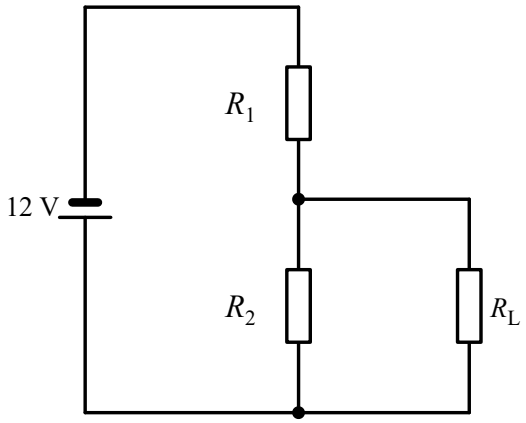
Two resistors (or a resistance wire with a sliding contact) in series across a supply share the supply p.d. in the ratio of their resistances 1

Changing the ratio (moving the slider) gives an output p.d. that can be varied from zero up to the full supply voltage 1

2

Explanation: $V_{out} = V_{in}R_2/(R_1 + R_2)$.

(b) A battery of 12.0 V and negligible internal resistance is connected to three resistors, R_1, R_2 and R_L, as shown in the diagram below.



If the resistances of the resistors R_1, R_2 and R_L are 5.0 kΩ, 10.0 kΩ and 5.0 kΩ respectively, calculate

(i) the output voltage V_{out} across R_L, [2]

Answer

$R_2 \parallel R_L = \frac{(10.0)(5.0)}{15.0} = 3.33$ kΩ — 1

$V_{out} = 12.0 \times \frac{3.33}{5.0 + 3.33} = 4.8$ V — 1

2

Explanation: The load reduces the output p.d. (it would be 8.0 V without R_L).

(ii) the current through R_1, R_2 and R_L, [5]

Answer

P.d. across $R_1 = 12.0 - 4.8 = 7.2$ V — 1

$I_1 = \frac{7.2}{5.0 \times 10^3} = 1.44$ mA — 1

$I_2 = \frac{4.8}{10.0 \times 10^3} = 0.48$ mA — 1

$I_L = \frac{4.8}{5.0 \times 10^3} = 0.96$ mA — 1

Check: $I_1 = I_2 + I_L$ — 1

5

Explanation: Kirchhoff's first law at the junction.

(iii) the power dissipated in R_L. [2]

Answer

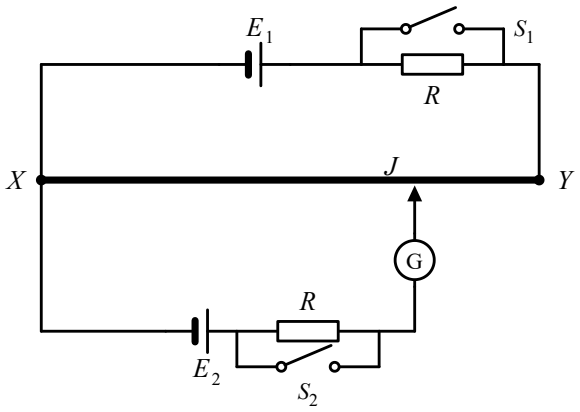
$P = I_L^2 R_L = (0.96 \times 10^{-3})^2 (5.0 \times 10^3)$ — 1

$P = 4.6 \times 10^{-3}$ W — 1

2

Explanation: Or $V^2/R = 4.8^2/5000$.

(c) A potentiometer circuit consists of a uniform slide wire XY, two cells with negligible internal resistance E_1 and E_2, two identical resistors R, two switches S_1 and S_2, and a galvanometer, as shown in the diagram below.



The length of XY is 100.0 cm with resistance 3.0 Ω. If the e.m.f.s of E_1 and E_2 are 2.0 V and 1.3 V respectively, and the resistance of R is 0.50 Ω, determine the balanced length of XJ when

(i) both S_1 and S_2 are closed, [2]

Answer

S_1 closed shorts out R in the driver circuit: $V_{XY} = 2.0$ V, gradient 0.020 V cm^{-1} 1

$XJ = \dfrac{1.3}{0.020} = 65.0$ cm 1

2

Explanation: At balance no current flows through E_2, so S_2 makes no difference.

(ii) S_1 is opened and S_2 is closed. [2]

Answer

S_1 open: $V_{XY} = 2.0 \times \dfrac{3.0}{3.0 + 0.50} = 1.71$ V 1

$XJ = 100 \times \dfrac{1.3}{1.71} = 75.8$ cm (about 76 cm) 1

2

Total for question **15**

Explanation: The extra resistance in the driver circuit lowers the potential gradient, so the balance length increases.